SORTING COLOUR BLOCKS

Game Design Document

Version 1.0.0

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1. Overview

1.1 Pitch

Sorting Colour Blocks is a relaxing yet challenging mobile puzzle game where players sort coloured blocks into matching lanes.

1.2 Genre

Puzzle, Casual.

1.3 Theme

Colorful and satisfying.

1.4 Target Platform & Audience

Platform	Android, WebGL
Audience	Casual players who enjoy puzzles
Age Range	All ages

1.5 Concept

A colour-sorting puzzle game where players move blocks between lanes to group matching colours and clear each level. Short sessions, escalating challenge, no timers.

2. Core Gameplay

2.1 Core Loop

- Load a level
- Observe the current block arrangement
- Move blocks between lanes by tapping
- Use Undo or Hint if needed
- Solve the puzzle
- Earn stars and unlock the next level

3. Game Systems

Block Movement

- Tap a lane to select the top block.
- Tap a destination lane to move it.
- A move is valid only if the destination lane is empty or the top block matches the moving blocks colour.
- Each lane has a maximum block capacity.

Win Condition

- A level is cleared when every lane is fully sorted by its color or some of the lanes are empty.
- Completed lanes are locked automatically.

Star System

- 3 Stars: completed within ideal moves.
- 2 Stars: completed within ideal+ 5 moves.
- 1 Star: completed in more moves than the ideal moves + 5.

Undo System

- Players can undo the most recent move.
- Undo supports experiment from players to solve the puzzle.
- Already available from level 1.

Hint System

- The game suggests a valid next move using a BFS pathfinder.
- Each level starts with 3 of hints.
- The source lane flashes yellow and the destination lane flashes green.
- Already available from level 1.

Puzzle Generation

- Levels are built from predefined configurations (colours, blocks, empty lanes).
- The board is scrambled into a playable state using random valid moves.
- A BFS solvability check confirms the puzzle can be completed.
- If unsolvable, the generator retries up to 5 times automatically.

Progression and Save System

- Stars earned per level are saved using PlayerPrefs.
- Completing a level unlocks the next.
- The Level Select screen shows each levels locked/unlocked and best star rating.
- Progress saved between sessions on both Android and WebGL (browser localStorage).

Tutorial

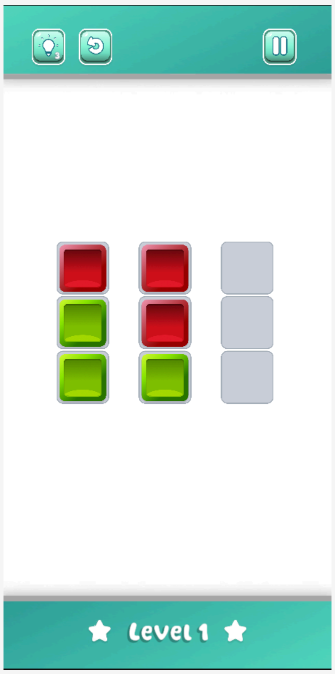
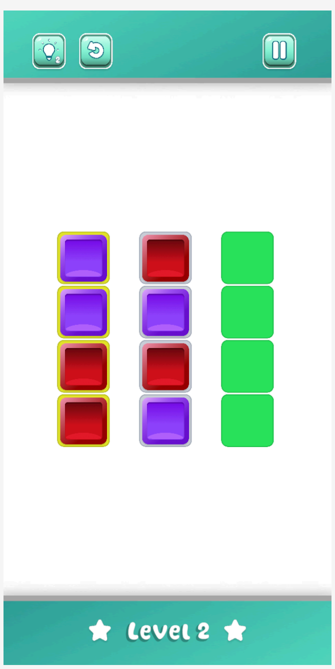
- A short one-time tutorial plays at the start of the first session.
- There is 4 steps inside the tutorial.
- A spotlight overlay darkens the screen and highlights the active element.
- The tutorial plays once and never repeats after it completes.

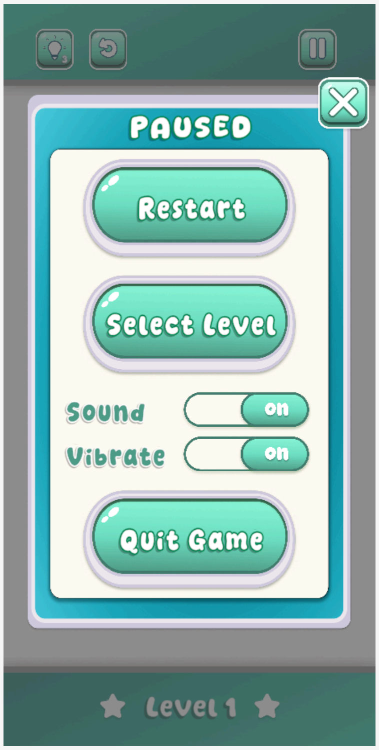
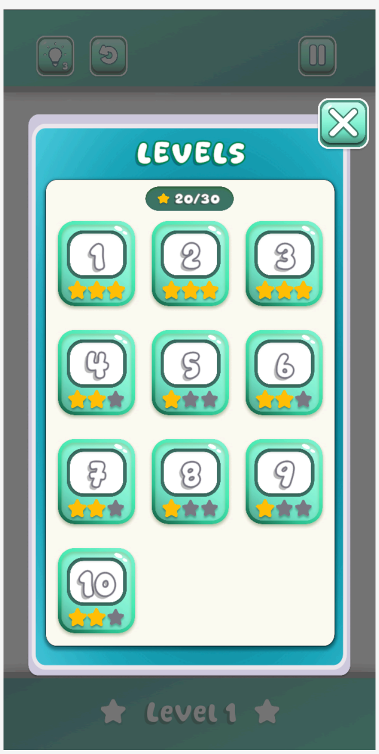
4. Level Design

The game contains 10 levels. Early levels introduce basic rules with few colours and generous empty lanes. Later levels increase difficulty with more colours, tighter lane space and higher ideal move requirements.

Level	Colors	Blocks/Color	Empty Lanes	Ideal Moves	Notes
1	2	3	1	5	Tutorial level
2	2	4	1	6	Longer blocks per colour
3	3	3	2	7	Introduces third colour
4	2	5	2	8	More blocks per colour
5	3	4	1	9	Fewer empty lanes
6	4	3	2	10	Four colours introduced
7	4	4	1	11	Fewer empty lanes
8	5	4	1	15	Five colours introduced
9	4	6	1	16	Six blocks per colour
10	5	8	1	20	Maximum challenge

5. UI Design

#	Screen
1	Gameplay - level 1 
2	Hint Button and Highlight 

3	Pause Menu 
4	Select Level Panel 

5	Win Panel 

5. Controls

Action	Input
Select a block	Tap a lane
Move a block	Tap a destination lane after selecting
Undo last move	Tap the Undo button
Get a hint	Tap the Hint button
Pause	Tap the Pause button

6. Technical Notes

Engine	Unity (2D)
Language	C#
Save System	Unity PlayerPrefs (Android & WebGL)
Input	Unity Input System (touch + mouse)
Board System	Level prefab
Puzzle Logic	BFS solvability check, scramble validation
UI	Unity UI + TextMeshPro
Design Tool	Figma (UI & assets)
Audio	AudioManager with sound and vibration toggles